

Java Interview Question Bank

250+ questions across 18 topics — Spring Boot, Microservices, Kafka, DSA, System Design & more

frequently asked — Almost always comes up

must know — Know this cold

concept — Good to know

Java Core (40 questions)

Always asked — OOP, collections, multithreading, Java 8+

OOP & FUNDAMENTALS

must know	What is the difference between abstract class and interface? When to use which?
must know	Explain polymorphism with a real example from your project.
must know	What is method overloading vs overriding?
concept	What is the diamond problem? How does Java solve it?
must know	What are the four pillars of OOP? Explain each with examples.
frequently asked	What is the difference between composition and inheritance? Which is preferred?
must know	What is static keyword? Can a static method be overridden?
must know	What is final keyword in Java — for class, method, and variable?

COLLECTIONS & GENERICS

frequently asked	HashMap internals — how does it store and retrieve values? What happens on hash collision?
must know	Difference between HashMap, LinkedHashMap, and TreeMap.
must know	ArrayList vs LinkedList — which is faster for get() and add()?
frequently asked	What is ConcurrentHashMap? How is it different from HashMap?
must know	What is the difference between Comparable and Comparator?
concept	What is fail-fast vs fail-safe iterator?
must know	What is the difference between HashSet and TreeSet?
concept	What is a PriorityQueue? How does it work internally?
concept	What is a WeakHashMap? When would you use it?

MULTITHREADING & CONCURRENCY

must know	What is the difference between Thread and Runnable?
must know	What is synchronized keyword? How does it prevent race conditions?
frequently asked	What is volatile keyword? When do you use it?
frequently asked	What is a deadlock? How do you avoid it?
must know	What is ExecutorService? Why use it instead of creating threads directly?
frequently asked	Explain Future and CompletableFuture — what's the difference?
must know	What is the difference between wait() and sleep()?
must know	What is a thread pool? What are the different types?
frequently asked	What is a race condition? Give an example.
concept	What is ReentrantLock? How is it different from synchronized?

JAVA 8+

must know	What are functional interfaces? Give 3 examples.
frequently asked	Explain Stream API — filter, map, reduce with examples.
must know	What is Optional? Why was it introduced?
frequently asked	What is the difference between map() and flatMap() in streams?
must know	What are default methods in interfaces? Why were they added?
frequently asked	What is a method reference? Give 4 types with examples.
concept	What is the difference between stream() and parallelStream()?
concept	What are records in Java 16+?

EXCEPTION HANDLING

must know	What is the difference between checked and unchecked exceptions?
must know	What is the difference between throw and throws?
concept	Can finally block override a return statement?
must know	What is try-with-resources? When to use it?
concept	What is exception chaining?

JVM Internals (14 questions)

Comes up at 5+ years experience — [show depth here](#)

MEMORY MODEL

must know	What is heap vs stack memory in JVM?
frequently asked	What is the difference between Young generation and Old generation?
concept	What is PermGen vs Metaspace?
frequently asked	What causes a memory leak in Java? How do you find it?
frequently asked	What is OutOfMemoryError? What are the different types?
must know	What is a StackOverflowError? When does it happen?

GARBAGE COLLECTION

must know	How does garbage collection work in Java?
concept	What is the difference between G1GC, ZGC, and Serial GC?
concept	What is Stop-the-World pause?
must know	What is the difference between Minor GC and Major GC?
frequently asked	How do you tune JVM heap size? What are -Xms and -Xmx?

CLASS LOADING

concept	What is class loading in JVM? What are the 3 types of class loaders?
must know	What is the difference between JVM, JRE, and JDK?
concept	What is bytecode? How does JIT compiler work?

Spring Boot (28 questions)

Core of every Java interview — know these deeply

CORE CONCEPTS

frequently asked	What is Spring Boot auto-configuration? How does it work internally?
must know	Explain @SpringBootApplication — what annotations does it combine?
must know	What is the difference between @Component, @Service, @Repository, @Controller?
must know	What is dependency injection? What are the 3 types in Spring?
frequently asked	What is the Spring Bean lifecycle?
frequently asked	What is the difference between @Bean and @Component?
must know	What is @Primary and @Qualifier?

must know	What is @Scope? What are the bean scopes in Spring?
concept	What is ApplicationContext vs BeanFactory?
must know	What is @Profile? How do you switch between dev and prod?

SPRING DATA & JPA

frequently asked	What is the difference between EAGER and LAZY loading?
frequently asked	What is the N+1 problem? How do you solve it?
must know	What is @Transactional? What happens if it's missing?
must know	Explain Spring Data JPA vs Hibernate vs JDBC.
frequently asked	What is optimistic vs pessimistic locking in JPA?
must know	What is @OneToMany and @ManyToOne? What is the owning side?
must know	What is JPQL vs native query?

SPRING SECURITY

frequently asked	How does Spring Security filter chain work?
must know	What is JWT? How do you validate it in Spring Security?
must know	What is the difference between authentication and authorization?
frequently asked	How does your JWT filter work across services?
must know	What is CSRF? Why disable it for REST APIs?
must know	What is CORS? How do you configure it in Spring Boot?
must know	What is @PreAuthorize and @PostAuthorize?

SPRING CLOUD

must know	What is Eureka? How does service registration and discovery work?
frequently asked	What is Feign Client? How is it different from RestTemplate?
frequently asked	What is Resilience4j? What is a circuit breaker pattern?
must know	What is load balancing in Spring Cloud? How does it work?

Microservices (14 questions)

Your project is your strongest asset — answer with real examples

DESIGN PRINCIPLES

frequently asked	Why did you split into microservices? What were the trade-offs?
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frequently asked	How do your services communicate — sync vs async? Why?
must know	What is service discovery? How does Eureka work in your project?
must know	What is an API Gateway? What does your gateway do?
must know	What is the difference between monolith and microservices?
concept	What is domain-driven design (DDD)?

RELIABILITY & RESILIENCE

frequently asked	What happens if transaction-service is down during a deposit?
frequently asked	What is a circuit breaker? Have you implemented one?
must know	What is eventual consistency? How does it apply to your system?
frequently asked	How do you handle distributed transactions — 2PC vs Saga pattern?
frequently asked	What is a retry mechanism? What is exponential backoff?

DATA MANAGEMENT

frequently asked	Why does each microservice have its own database?
concept	What is the CQRS pattern?
frequently asked	How do you handle data consistency across services?

Kafka (15 questions)

This separates you from most candidates — know it deeply

CORE CONCEPTS

frequently asked	Why did you use Kafka instead of REST calls between services?
must know	What is a topic, partition, offset, and consumer group?
frequently asked	What guarantees does Kafka give — at most once, at least once, exactly once?
frequently asked	What happens when a Kafka consumer fails mid-processing?
frequently asked	What is a Kafka producer acknowledgment (acks=0, 1, all)?
must know	What is consumer lag? How do you monitor it?
frequently asked	What is the difference between Kafka and RabbitMQ?
frequently asked	What is a dead letter queue (DLQ) in Kafka?

PARTITIONING & ORDERING

frequently asked	How does Kafka decide which partition a message goes to?
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frequently asked	How do you guarantee message ordering in Kafka?
must know	How do you scale a Kafka consumer group?

IN YOUR PROJECT

frequently asked	Walk me through a deposit event — what happens step by step?
must know	What Kafka topics do you have and why?
frequently asked	How would you handle duplicate messages in Kafka?
frequently asked	What is idempotent consumer? How would you implement it?

Redis (12 questions)

Rate limiting + caching — you used it, so own this topic

CORE CONCEPTS

must know	What is Redis? Why is it faster than a relational database?
must know	What data structures does Redis support? When to use each?
frequently asked	What is cache eviction? What are the different eviction policies?
frequently asked	What is a cache stampede? How do you prevent it?
concept	What is Redis persistence — RDB vs AOF?

CACHING PATTERNS

frequently asked	What is cache-aside (lazy loading) pattern?
frequently asked	What is write-through vs write-behind caching?
frequently asked	How do you handle cache invalidation? Why is it hard?
frequently asked	What TTL would you set for a bank account cache? Why?

IN YOUR PROJECT

frequently asked	How does Redis rate limiting work in your gateway?
frequently asked	What is the token bucket algorithm for rate limiting?
frequently asked	What would happen if Redis goes down — what's your fallback?

Databases (16 questions)

SQL + indexing + transactions — every company tests this

SQL & JOINS

must know	What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN?
frequently asked	What is a CTE (Common Table Expression)? Give an example.
must know	What is the difference between HAVING and WHERE?
frequently asked	What is a window function? Give an example with RANK().
frequently asked	Write a query to find the second highest salary.
frequently asked	Write a query to find duplicate records in a table.

INDEXES & PERFORMANCE

must know	What is an index? How does it speed up queries?
frequently asked	What is the difference between clustered and non-clustered index?
frequently asked	What is a composite index? When does column order matter?
frequently asked	What is EXPLAIN / EXPLAIN ANALYZE in PostgreSQL?
frequently asked	When should you NOT add an index?

TRANSACTIONS & ACID

must know	What are ACID properties? Explain each with an example.
frequently asked	What is isolation level? What are the 4 levels?
frequently asked	What is a dirty read, phantom read, and non-repeatable read?
frequently asked	What is a deadlock in a database? How do you prevent it?
must know	What is connection pooling? What does HikariCP do?

Docker (11 questions)

You built it — answer with your actual docker-compose experience

CORE CONCEPTS

must know	What is the difference between a Docker image and a container?
must know	What is a Dockerfile? Walk me through yours.
must know	What is Docker Compose? Why did you use it?
must know	What is the difference between CMD and ENTRYPOINT?
frequently asked	What is a Docker volume? How do you use it for PostgreSQL?
frequently asked	What is a Docker network? What are the different types?

frequently asked	What is multi-stage build in Docker? Why is it useful?
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IN YOUR PROJECT

frequently asked	How do your 6 services communicate inside Docker network?
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frequently asked	How do you handle service startup order in docker-compose?
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must know	What health checks do you have in your docker-compose?
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must know	How do you pass environment variables to Docker containers?
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Design Patterns (9 questions)

Tie every pattern to a real example from your project

CREATIONAL

frequently asked	What is the Singleton pattern? How do you make it thread-safe?
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must know	What is the Factory pattern? When do you use it?
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frequently asked	What is the Builder pattern? Where did you use it in your project?
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STRUCTURAL & BEHAVIORAL

must know	What is the Adapter pattern? Give a real example.
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frequently asked	What is the Proxy pattern? How does Spring AOP use it?
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frequently asked	What is the Observer pattern? How does Spring Events use it?
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must know	What is the Strategy pattern? Give an example.
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frequently asked	What is the Chain of Responsibility pattern? How does Spring Security use it?
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frequently asked	Which design patterns did you use in your project and where?
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Testing (10 questions)

You have 88+ tests — this is your strength, own it

UNIT TESTING

must know	What is the difference between unit testing and integration testing?
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must know	What is Mockito? What is the difference between @Mock and @InjectMocks?
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frequently asked	What is the difference between @MockBean and @Mock?
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must know	What is verify() in Mockito? When do you use it?
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frequently asked	What is argument captor in Mockito?
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SPRING TESTING

must know	What is @SpringBootTest? When do you use it?
frequently asked	What is @WebMvcTest? What does it load?
must know	What is MockMvc? How do you test a REST endpoint?
frequently asked	What is TDD? How did you apply it in your project?
frequently asked	How do you test Kafka consumers in your project?

REST & API Security (11 questions)

Basics every backend developer must know cold

REST PRINCIPLES

must know	What are the constraints of REST architecture?
frequently asked	What is idempotency? Which HTTP methods are idempotent?
must know	What is the difference between PUT and PATCH?
must know	What HTTP status codes do you use and when?
frequently asked	How do you version a REST API?
must know	How do you design pagination in a REST API?
frequently asked	What is API idempotency key? How do you implement it for payments?

SECURITY

frequently asked	What is OAuth 2.0? How is it different from JWT?
must know	What is an access token vs refresh token?
must know	What is SQL injection? How do you prevent it in Spring Boot?
must know	What is XSS? How do you prevent it?

Logging & Observability (11 questions)

Production mindset — shows you think beyond just making it work

LOGGING

must know	What are the log levels? When do you use each?
frequently asked	What is structured logging? Why is it better than plain text logs?
frequently asked	What is a correlation ID? How do you pass it across microservices?
frequently asked	How do you centralize logs from 6 microservices?

frequently asked	How do you avoid logging sensitive data like passwords or card numbers?
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METRICS & TRACING

must know	What is Spring Boot Actuator? What endpoints does it expose?
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must know	What is Micrometer? How does it work with Prometheus?
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frequently asked	What metrics would you monitor for a payment service?
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must know	What is distributed tracing? Why is it needed for microservices?
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must know	What is Zipkin? What is a trace ID and span ID?
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frequently asked	How would you use Zipkin to debug a slow deposit request?
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Git & DevOps (9 questions)

Often skipped in prep — don't get caught out

GIT

must know	What is the difference between git merge and git rebase?
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must know	How do you resolve a merge conflict?
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frequently asked	What is GitFlow branching strategy?
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frequently asked	What is the difference between git reset and git revert?
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must know	What is a pull request / code review process?
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CI/CD

must know	What is CI/CD? What is the difference between them?
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must know	What is a build pipeline? What stages does it have?
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concept	What is blue-green deployment?
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concept	What is canary deployment?
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DSA (17 questions)

Product companies always include DSA rounds — practice daily

ARRAYS & STRINGS

frequently asked	Find the two numbers in an array that add up to a target. (Two Sum)
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frequently asked	Find the longest substring without repeating characters.
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must know	Rotate an array by K positions.
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frequently asked	Find the maximum subarray sum. (Kadane's algorithm)
must know	Merge two sorted arrays without extra space.

HASHMAP PROBLEMS

frequently asked	Find the first non-repeating character in a string.
frequently asked	Group anagrams from an array of strings.
frequently asked	Find the top K frequent elements.
must know	Check if two strings are anagrams.

LINKED LIST & TREES

frequently asked	Reverse a linked list iteratively and recursively.
frequently asked	Detect a cycle in a linked list. (Floyd's algorithm)
frequently asked	Level-order traversal of a binary tree (BFS).
must know	Validate a binary search tree.
concept	Find the lowest common ancestor of two nodes.

SLIDING WINDOW

frequently asked	Maximum sum of subarray of size K.
must know	Move all zeros to end of array.
frequently asked	Container with most water.

System Design (11 questions)

Think out loud, draw components, ask clarifying questions

COMMON QUESTIONS

frequently asked	Design a payment system like Razorpay or PhonePe.
frequently asked	Design a rate limiter. What algorithms would you use?
frequently asked	How would you scale your banking app to 10 million users?
must know	Design a notification system that handles email, SMS, and push.
must know	Design a URL shortener.
concept	Design a distributed cache.

CONCEPTS TO KNOW

frequently asked	What is CAP theorem? How does it apply to your system?
must know	What is horizontal vs vertical scaling?
must know	What is a message queue vs event streaming?
frequently asked	What is consistent hashing? Why is it used in distributed systems?
must know	What is a load balancer? What are the algorithms?

HR & Behavioral (10 questions)

Bad answers here cancel good technical answers — prepare these

ABOUT YOU

must know	Tell me about yourself. (2-minute structured answer)
must know	Why are you looking to change jobs?
must know	Where do you see yourself in 3-5 years?
frequently asked	What is your expected CTC?
frequently asked	Why do you want to join this company specifically?

SITUATIONAL

frequently asked	Tell me about a challenging bug you fixed. How did you approach it?
frequently asked	Tell me about a time you disagreed with your team. What happened?
must know	Describe a project you're most proud of and why.
frequently asked	Tell me about a failure. What did you learn?
must know	How do you handle tight deadlines?

Your Project (12 questions)

Spend the most time here — 30 min of every interview will be on this

ARCHITECTURE

must know	Walk me through your entire system architecture end to end.
frequently asked	Why 6 services? What would you merge or split differently?
must know	How does the full deposit flow work end to end — every step?
must know	How does JWT authentication work across all your services?
frequently asked	What happens if Kafka is down — what's your fallback?

must know

Why Redis for rate limiting and not a database table?

IMPROVEMENTS & TRADE-OFFS

frequently asked

What was the hardest debugging session you had?

frequently asked

What would you do differently if you started over?

frequently asked

How would you make this production-ready?

frequently asked

How would you handle 100k concurrent users on this system?

frequently asked

What security vulnerabilities might exist in your current design?

concept

How would you deploy this to AWS?